

CARL MATHEW ESTORGA

COMPUTER ENGINEER

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SUMMARY

Computer Engineering graduate with a passion for programming and software maintenance, backed by a strong foundation in circuit diagram and flowchart design. Knowledgeable in Python, HTML, and CSS, with hands-on experience in microcontroller programming, circuit analysis, and software troubleshooting. Proficient in Microsoft 365 and Google Workspace for efficient documentation, data management, and collaborative work.

TECHNICAL SKILLS

- Raspberry Pi
- Linux OS
- Python
- SQL
- Microsoft 365
- Google Workspace

SOFT SKILLS

- Collaborative
- Organized
- Adaptable
- Approachable
- Active Listener
- Time Management
- Problem-Solving

PROFESSIONAL EXPERIENCE

Computer Engineering Intern
University of Santo Tomas - Legazpi

February 2025 - May 2025

During my internship at the University of Santo Tomas - Legazpi, I experienced what it is like to work as an office staff member and as a technician. The notable skills I developed during my internship include professional communication and computer maintenance. I was also taught advanced features of Microsoft 365 apps, such as MS Word's mail merge and field braces, and MS Excel formulas. Additionally, I worked on projects such as creating a daily time record system that automatically calculates the hours spent on the internship using time-in and time-out columns. Lastly, although not explored in depth during my internship, I was also exposed to setting up computer networks, including installing a new server and troubleshooting a faulty router.

EDUCATION

University of Santo Tomas-Legazpi

Finished 2025

- Bachelor of Science in Computer Engineering.
- Thesis: AI Camera-Integrated Waste Collection and Segregation System Using Raspberry Pi – designed and implemented a system capable of automated waste identification, collection, and sorting.
- Developed strong skills in creating clear flowcharts, logic circuits, and circuit diagrams for system planning and execution.

PROJECTS

I. Scientific Calculator App

Developed a virtual CASIO fx-100MS calculator using MATLAB App Designer, capable of mimicking the actual calculator such as basic arithmetic operations, number system conversions (binary, octal, hexadecimal), and sin/alpha mode functions.

II. Store Inventory & Records App

Developed a desktop application using Python, PyQt6, and SQLite database to manage store operations, including product stock tracking, customer payment records, and a daily income tracker.

III. File Organizer App

Developed a Python desktop application with PyQt6 that automatically organizes files into a generated folder. The app supports two modes: (1) General organization, which sorts files by category, and (2) Name-based organization, which strictly organizes documents, images, and records by filename.

IV. Garbage Collecting & Segregating Robot

Trained a PyTorch model using the YOLOv5 Framework and developed an automated garbage collecting robot that also segregates the trash based on the camera reading using Raspberry Pi and Python as the programming language.

V. Obstacle Avoidance Robot

Developed an Arduino-based obstacle-avoiding robot with a team, utilizing an ultrasonic sensor for real-time object detection and autonomous navigation.

CERTIFICATES

- **Responsive Web Design** – FreeCodeCamp, July 2025
- **Certificate of Completion** – University of Santo Tomas–Legazpi Educational Technology Center, May 2025
- **Setting up computer networks** – Technical Education and Skills Development Authority (TESDA). July 2024
- **AI Revolution Unlocking the Potential of Tomorrow** – Institute of Computer Engineers of the Philippines, June 2024
- **Introduction to Python** – SoloLearn, August 2023

ADDITIONAL INFORMATION

- **Languages:** Tagalog, Bikol, English.
- **Address :** 45 Loyola St., Purok 2 Barangay IV (Poblacion), Malilipot, Albay